

Estimation of Varietal Classification and Quality Analysis in Rice Seeds Using Digital Image Processing – A Survey

M. Sandhiya
Research Scholar
Saveetha School of Engineering,
Saveetha Institute of Medical and
Technical Sciences,
 Chennai, India
 288sandhyajes@gmail.com

Dr. S. Christy
Associate Professor
Saveetha School of Engineering,
Saveetha Institute of Medical and
Technical Sciences,
 Chennai, India
 christy.s@saveetha.edu.in

Abstract— Agriculture plays a major role in today’s world. 70% of rice production is cultivated in south India. Thailand, the US, India, Vietnam, etc., are exporting rice all over the world. But they are facing one of the most difficult tasks: assuring the rice seed quality. Good seeds can increase production, so selecting seeds is very important. A maximum of the work was done physically or by using huge, expensive machines. For choosing good seeds, agronomists be conscious of two factors, such as the variety which is determined by some quality criteria, and next by the skilled one who is going to analyze the rice seed physically. Both tasks take a long time and the result of accuracy is also not as expected. Nowadays computer vision technology with Digital Image Processing plays a major role in many fields, namely agronomy, healthcare, machine vision, pattern recognition, remote sensing, Video processing, etc. Our research work mainly focuses on the agricultural arena, going to provide good quality seeds and identify dissimilar varieties of rice seeds by digital image processing techniques such as image capturing, segmenting the rice seeds, extracting the features from the rice seeds and finally feeding into the machine learning algorithm to find out the accuracy too.

Keywords—*Machine vision, Rice seed varietal purity, Classification of rice seeds, Quality analysis of rice seed, Machine learning algorithm.*

I. INTRODUCTION

The second-largest production of rice, wheat, cotton, groundnuts, vegetables, and fruits are produced in India. Rice is the oldest and most widely grown cereal grain. 90 % of rice is cultivated in south India. Increasing the growth of the population, there is a demand for rice. At the same time, the rice production rate is reduced due to water demand, metrological conditions, insect bouts, crop adulterations, reduced amount of germination plants, diseases, less quality fertilizer used to the plant, etc.

The main objective of this research work focuses on choosing quality seeds and good seeds lead to better production.

The yield of the grain is determined by the quality of the rice seeds, infertility of the rice seed not only affects yield, it also reduces the market rate of the product. So, for all rice makers quality testing is the best practice for selecting good seeds. Due to the fact that physical testing is based on what they sense and observe, it takes a long time and there is no assurance of correctness. Analyze the physical features of the seed, such as form, size, color, length, and width, using a computer system.

Figure 1 depicts the formation for the production of rice seeds. One of the essential and most crucial duties is choosing the right seed variety and determining its quality [2]. Because seed is the base of every rice crop and is essential to production, it is crucial that it be correctly identified, collected, and harvested to maximize yield and quality.

Seed classification criteria

- Pure Variety
- Seed Viability
- Moisture level



Fig. 1. Rice’s life cycle.

A. The following conditions must be met for a variety to be pure

- Red rice seed
- Inert materials (stones, soil, etc.)
- Unwanted seeds and other harvested seeds
- Other mixed varieties
- Moisture content
- Germination

$$\text{Germination \%} = (\text{No. of Seeds germinated}) / (\text{No. of seeds taken for the sample})$$

- Calculation of unwanted Crop seeds and

$$\text{Weed \%} = (\text{weight of weed}) / (\text{Weight of seeds taken for the sample})$$

- Calculation of unwanted items such as gravel, dirt, soil, etc.

$$\text{Unwanted materials \%} = (\text{Weight of Unwanted materials}) / (\text{Weight of seeds taken for the sample})$$

B. The following variables are used to determine seed viability

The germination process, force, and moisture level of the seeds are used to determine their viability. The technique of determining whether or not seeds are alive is known as seed viability. While some seeds are viable for many years, some are not.

C. Calculation of humidity content in the rice seed

The water content present in the paddy serves as a gauge for the humidity content of the grain. A good seed is one that is capable of germination and has a water content of less than 14 percent. A moisture meter is typically used to measure the moisture content of rice seeds since it provides better and faster results but only works with tiny samples. Capacitive moisture meters are required for measuring big samples, but they are costly.

D. Rice seed quality testing equipment

For analyzing the viability of seeds, the moisture level in rice seeds, and variety purity for germination, IIRI created an assessment kit. Each component's performance is limited to one particular test and is too expensive.

- Rice Milling Chart
- IIRI Moisture Tester
- Magnifier
- Indented Sheet Grader
- Scale
- Calliper,
- Graduated cylinder
- Infrared Thermometer
-

Figure 2 displays the IIRI Rice Knowledge Bank's equipment for evaluating the quality of rice seeds



Fig.2. IIRI Tool kit

II. BASIC HARDWARE SETUP REQUIREMENTS

Due to the high cost of the IIRI Rice seed quality assessment kit, we are switching to a computer-based machine vision technique to evaluate the rice seed's quality utilizing the aforementioned variables. Figure 3 includes the following

- A illumination device that illuminates the samples being tested.
- A solid-state CCD camera to capture images.
- A high-resolution color display,
- A frame-grabber
- A personal computer

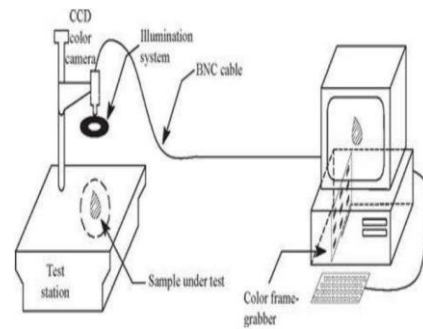


Fig. 3. Typical Hardware Configuration to Gather Sample Data.

III. EXPERIMENTAL RESULT

A. Variety of rice seed samples

[1] They proposed a digital image analysis algorithm for finding the rice seed variety. They used the following six seed types: syz3, xs11, xy5968, ey7954, xy9308, and z903. From these, they extracted 6 color features and 13 morphological features. Area 1 (mm²), Area 2, Thinness ratio, Equivalent diameter, length, breadth, rectangular aspect ratio, Major axis length, Minor axis length, filled area, convex area, solidity, and extent are the morphological features taken in this paper. Red (R), green (G), blue (B), hue (H), saturation (S), and intensity (I) are the six color features they find.

[2] In this paper they look at the rice seed variation, they identified them using reversed neural network. They took 5 different rice seed varieties Xiannong, Jinyougui, You166, Xiannong, and Medium you with an accuracy of 99.99%, 99.93%, 98.89%, 82.82%, and 86.65% respectively.

[3] This paper introduces Computer vision and image processing techniques to analyze different rice seed varieties. They collected six types of commonly grown rice in northern Vietnam: Huang thom 1, BC-15, Thien_uu, Nep-87, and Q-5. They extracted 8 morphological features, 3 color features, and 4 texture features. Then using Random forest, SVM, and KNN classifier the extracted features are classified and provide 90.54% accuracy.

[4,5] The four separate paddy varieties: R-2, R-4, K-6and R-24 can be classified using an artificial neural network, according to a proposal made by Archana Chaugule and Suresh N. Mali. They extracted texture, color, and Shape features from the image and calculated the accuracy of each 82.61%, 88.00%, and 87.27%.

[6,7] The authors identify non-destructive pure-breeding rice seeds using digital image processing techniques. They used the rice seed named, Khao Dawk Mali 105 for analysis. The classification was made by segmented image and RGB color features and got the accuracy rate for healthy rice seeds was 98% and 82% for pure breeding rice seeds.

[8,9] This paper used a feed-forward Neural Network to categorize five dissimilar varieties of rice kernels namely Fjr, Neda, Hashemi, Mahali, and Gerde. Eighteen morphological features are extracted from the image due to redundant, noisy, and irrelevant information they used four different feature selection algorithms as a standard backward sequential, branch and bound, plus-1-takeaway-r,

Standard sequential forward, and finalized 8 morphological features and obtained the accuracies were 100%, 100%, 96%, 100%, 96%.

B. Image Acquisition

[1] The images were captured with a color CCD camera (Model TMC7DSP, Pulnix) with a 640 x 480-pixel resolution. The field of view was 12 mm by 9 mm, and the spatial resolution was around 0.019 mm/pixel. All photographs taken of rice seedlings have been verified by people. A whiteboard served as the background.

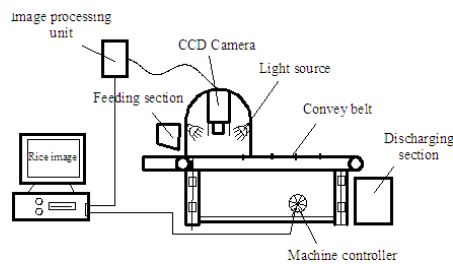


Fig. 4. Automatic Rice Seed Variety System.

[2] Figure 4. displays A G. OuYang's suggested technique. The rice seeds were emptied from the container and scattered on a predetermined lattice that moved over the belt during the photo-taking process. CCD cameras take pictures of the seed-filled moving belt that is under the camera. The cameras are already connected to a computer to save photos. And the computer's photo preparation unit received the captured images. They selected a black background and a pale green surface color.

[3] The 640 x 480-pixel image was taken by the authors using a color CMOS image sensor camera (NIKON D300S). A room with a background of a white table. The rice seeds are dispersed over a 10 x 16 cm² space. They used digital cameras to acquire the images from different distances. The background was set up in black color. The images are fixed correctly based on the adjustment of the camera.

[4] They used an 18.9-megapixel computerized camera from Sony, the background was set up with a sheet of black material, and images of rice seeds. For testing, seeds were gathered from the seed testing facility in Pune, India. Four paddy grains' unclear business tests—R2, K6, R24, and R2—were gathered.

[5] S. Khunkhett, T. Remsungnen, et al.[5] used a scanner called (canon model MP287, 1200 X 2400 dpi) to capture the images. The photos are then delivered straight to the Compaq Model Presario V3000 Intel Pentium Dual-Core PC.

[6] S.J. MousaviRad, F. Akhlaghian Tab, and K.Mollazade used a flatbed scanner to acquire the images of rice kernels with 8-bit grayscale and a resolution of 600 dpi. The measurements were made for 300 rice kernels of each cultivar. The background was black color to extract the kernels easily.

C. Image Segmentation

The most important and fundamental step in the processing of digital images was image segmentation. Until we have a clear or segmented image for testing, we will partition the image into a number of components.

[1,10] In this study, threshold values are created using the histogram analysis results. This revealed that the blue value between the backdrop and the items was varied. Under VC++6.0, the threshold value is computed using the RGB color model. In this study, 200 was chosen as the threshold value. All pixels with a blue value of more than 200 were given a value of 0, and any operation on pixels with a blue value of 200 or less was skipped.



Fig. 5. a) The Original rice seed image of ey7954



Fig. 5. b) Image after Segmentation

[2] A double image is often the result of division. There are only two types of pixels in a two-dimensional image: those with a dark level estimation of 0 or 1. At a dark level of 1, objects of interest are detached in pixel gatherings, which are still rudimentary information needing an appropriate representation for future PC manipulation. The segmented sample is displayed in Figure 5 a & b.

[3] In this paper they fixed a threshold value has 90. The intensity in blue color is high, Therefore, all blue pixels were given the value 0, and any blue pixels that had a value lower than 90 were given the value 255 which is shown in Figure 6.

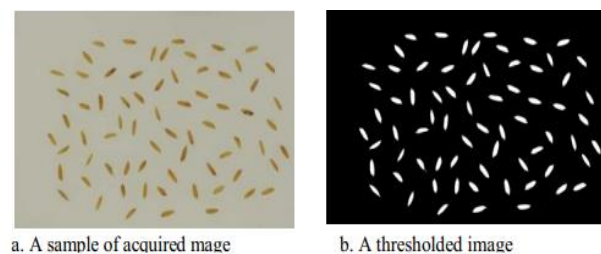


Fig. 6. An illustration of a segmented image that was collected.

[5] They used segmentation fixing the threshold value based on the histogram analysis of the rice image. An illustration of a segmented image that was collected. They set a threshold value of 110 in this paper. Every pixel with a grey value of less than 110 was given the value 0, while every pixel with a grey value of 110 or higher was left

unprocessed. The background was set to level 0, and the region containing the rice kernels was set to level 0.

[6] The authors segmented the image using edge detection operation at the interior of the rice seed. Then the original rice image is converted into RGB into grayscale image then converted into black and white(Binary image). Fig 7 shows the Binary Image.



Fig. 7. Binary Image

D. Feature Extraction

[1] Separate rice seeds' colors and morphological characteristics were extracted by the author Liu Zhao-yan, Cheng Fang, Ying Yi-bin, and Rao Xiu-qin. Using MATLAB 6.5 they extract the morphological features and color features such as Major axis length (mm), Minor axis length (mm), Length (mm), Width (mm), Thinness ratio, Aspect ratio, Rectangular aspect ratio, Equivalent diameter, Convex area (mm²), area 2 (mm²), Convex area (mm²), solidity, extent, Red(R), Green(G), and Blue(B). Every characteristic carries out a specific task to assess the quality of each rice seed. [2] Using the programming language Visual C++ 6.0, they extracted the picture of rice seed and examined it using RGB (Red, Green, and Blue) color characteristics. For each inquiry, the Matrix onboard factual function was invoked to process the pixel zone and count the pixels. The ratio of the sound zone to the overall area was used to determine the main component (x). The second component (y) was the ratio of the sound region to the area of the neighboring developing organism segment. If the guess of the fetus part was accurate, the second component was the more crucial. The arrangement made extensive use of the two highlights. [3] In this work, the authors investigated four features namely global features and local features. GIST refers to morphological features and SIFT refers to local features. Morphological features namely color, and texture. Using a Gabor filter bank applied to the image to extract the features in the rice seed. [4] In order to analyze the rice seed quality, they extracted the texture feature. The texture is made up of texels, or texture natives, which are adjacent groups of pixels with some tonal and regional properties. Texture can be described by the power (shading) properties of texels and the organization of their spatial relationships. Scale is significantly subservient to textures. The language and measurable/auxiliary aspects of texture are ranked. When conducting a true texture analysis, texture highlights are created by processing the actual distribution of powers in various sections of the image in relation to one another. First-arrange, second-arrange, and higher-arrange measures are categorized based on the number of power focuses (pixels) in each mix. [5] S. Khun Khet and T. Remsungnen suggested employing digital image processing to identify non-destructive pure-breeding rice seeds. They used the rice seed from Khao Dawk Mali 105 for analysis. The segmented image and RGB color

features were used for the classification, which had an accuracy of 98% for healthy rice seeds and 82% for pure breeding rice seeds. [6] They used the plus-I-takeaway-r, branch and bound, standard backward sequential and standard forward sequential algorithms to extract 18 features from the rice seed picture, of which six were selected for further reference.

E. Classification

Random Forest

Random forest is an ensemble algorithm technique. RF is very efficient for working with large datasets. Using a single dataset we can construct N number of decision trees, based on the majority of votes the final decision tree was made. In the Random Forest algorithm model we need to mention the estimator count. Bagging and boosting are the two main techniques that were used in the Random Forest algorithm. In RF without removing any node we can able to build the highest extension of the tree at the same time every node is divided by using better subsets. The Random Forest Algorithm's main strengths are its ability to handle data sets with both continuous and categorical variables and its improved accuracy when handling classification issues.

Support Vector Machine

One of the most well-liked machine learning algorithms is the Support Vector machine. This algorithm deals with three-dimension array. A space splitting the datasets into certain groups based on their similarity of features, that occurrence of the line is called a hyperplane. Eq. (1) shows

$$W^T X + b = 0,$$

W^T – weight vector

b – bias

X – features vector

Each SVM feature is associated with a certain location's value. SVM will get good accuracy when the distance between the datasets and hyperplane is more. It follows low gamma and low regularization methods to provide a better result.

K – Nearest neighbors

This algorithm is simple, effective, and non-parametric and it is widely used in classification problems. Using Euclidean distance, we can predict the arrival of new dataset. In the KNN algorithm K value should not be very small or very high.

F. Result and Discussion

[1] Six rice seed varieties were used in the model's implementation, and the accuracy was 90.00%, 88.00%, 95.00%, 82.00%, 74.00%, and 80.00% for the varieties ey7954, syz3, xs11, xy5968, xy9308, and z903. Respectively. [2] In order to distinguish between different rice seed classes based on appearance, they employed a backward neural network. They took 5 different rice seeds varieties 'You166', 'Xiannong', 'Jinyougui', Medium you, and Xiannong, with an accuracy of 99.99%, 99.93%, 98.89%, 82.82%, and 86.65% respectively. [3] They

extracted 8 morphological features, 3 color features, and 4 texture features. Then using Random forest, SVM, and KNN classifier the extracted features are classified and provide 90.54% accuracy. [4] They extracted texture, color, and Shape features from the image and calculated the accuracy of each 82.61%, 88.00%, and 87.27%. [5] The classification was accomplished using RGB color characteristics and segment image data and calculated the accuracy of 98% for healthy rice seeds and 82% for pure breeding rice seeds. [6] Due to redundant, noisy, and irrelevant information, 18 morphological features were extracted from the image using four different feature selection algorithms: branch and bound, standard sequential forward, plus-1-takeaway-r, and standard sequential backward. Of the final 8 morphological features, the accuracy values were 100%, 100%, 96%, 100%, and 96%.

IV. CONCLUSION

In this paper, we gain knowledge about how to analyze good quality seeds in machine vision using the Digital Image Processing technique. Additionally, we spoke about the hardware configuration and specifications for each suggested method. For each suggested methodology, a thorough examination of the data collection process, image segmentation, and several feature extraction techniques is conducted. We talked about the many categorization techniques that are frequently used to categorize data sets using features that have been retrieved. Last but not least, we calculate the rate of accuracy they were able to accomplish for each suggested approach for the relevant rice seed variety. We strongly recommend replacing CCD camera photos with MRI or IR images while employing the same processes for greater accuracy.

REFERENCES

- [1] Liu, Zy., Cheng, F., Ying, Yb. et al. Identification of rice seed varieties using neural network. J Zhejiang Univ Sci B 6, 1095–1100(2005). <https://doi.org/10.1631/jzus.2005.B1095>.
- [2] A. -G. OuYang, R. -j. Gao, Y. -d. Liu, X. -d. Sun, Y. -y. Pan and Xiao-ling Dong, "An automatic method for identifying different variety of rice seeds using machine vision technology," 2010 Sixth International Conference on Natural Computation, Yantai, China, 2010, pp. 84-88, doi: 10.1109/ICNC.2010.5583370.
- [3] Phan Thi Thu Hong, Tran Thi Thanh Hai, Le Thi Lan, Vo Ta Ho- ang and Nguyen Thi Thuy "Identification Of Seeds Of Different Rice Varieties Using Image Processing And Computer Vision Techniques" J. Sci. & Devel. 2015, Vol. 13, No. 6: 1036-1042.
- [4] Archana Chaugule, Suresh N. Mali, "Evaluation of Texture and Shape Features for Classification of Four Paddy Varieties", Journal of Engineering, vol. 2014, Article ID 617263, 8 pages, 2014. <https://doi.org/10.1155/2014/617263>.
- [5] S. Khunkhett, T. Remsunngnen "Non-Destructive Identification Of Pure Breeding Rice Seed Using Digital Image Analysis" JICTEE-2014.
- [6] S. J. Mousavi Rad, F. Akhlaghian Tab, K. Mollazade . Design of an Expert System for Rice Kernel Identification using Optimal Morphological Features and Back Propagation Neural Network. International Journal of Applied Information Systems. 3, 2 (July 2012), 33-37. DOI=http://ijais12-450472.
- [7] A.K. Reshmy "Data Mining of Unstructured Big Data In Cloud Computing" International Journal of Business Intelligence and Data Mining, Vol.13 Issue 1-3 2017
- [8] L. Maria Michael Visuwasam "A Distributed Intelligent Mobile Application For Analyzing Travel Big Data Analytics". Journal Of Peer To Peer Networking And Applications, 2019
- [9] Devi, K , Muthusenthil, B, Deep Learning Based Security Model for Cloud based Task Scheduling,"KSII Transactions on Internet and Information", Vol.14, Issue 9,pp 3663-3679, Sept.2020
- [10] Hariharan B."A Hybrid Framework For Job Scheduling On Cloud Using Firefly And Bat Algorithm International Journal of Business Intelligence And Data Mining, Vol.15 No.4, 2019
- [11] B. Hariharan "Wbat Job Scheduler: A Multi-Objective Approach For Job Scheduling Problem On Cloud Computing" Journal Of Circuits, Systems And Computers, 29(6), 2018